

Coupled Epidemic and Economic Constraint Dynamics: A SER-SIR Framework with Market and Household Leakage Regimes

1. Introduction

Economic systems subjected to large-scale shocks do not respond solely through immediate price adjustments or output contraction. Instead, they exhibit a range of intermediate behaviors in which stress is accumulated, absorbed, redistributed, and ultimately released through different channels. The COVID-19 pandemic provides a clear example of such a shock, simultaneously affecting labor availability, mobility, consumption patterns, and sectoral interdependence. While existing analyses have largely focused on demand shocks, supply disruptions, and price dynamics, less attention has been given to how unresolved burden propagates through the system and where it ultimately exits.

This paper approaches the problem from a systems perspective by coupling an epidemic model with a sectoral economic state model. Epidemic dynamics are represented through a standard susceptible–infected–recovered (SIR) process, which provides a time-varying external forcing signal. Economic dynamics are represented using a Stress–Elasticity–Reaction (SER) framework, where stress captures accumulated burden, elasticity captures absorptive capacity, and reaction represents observable system response. The coupling allows epidemic pressure to enter the economic system as a structured, time-dependent input, rather than as a one-time shock.

Within this framework, the key question is not only how much stress is generated, but how that stress is routed once it exceeds local absorptive capacity. We introduce a decomposition of leaked stress into two distinct exit channels. The first, termed market-channel leakage, corresponds to stress that propagates through observable exchange mechanisms and institutional structures, appearing as measurable disruption or instability. The second, termed household-channel leakage, corresponds to stress that propagates through human, local, and temporal mechanisms—such as deferral, substitution, and reduced participation—that remain largely outside standard economic measurement.

The central result of the paper is that these two channels behave as competing exit routes for leaked stress. Across a broad parameter sweep of the coupled system, the two channels are consistently negatively related, with dominance switching across a narrow region in stress–elasticity space. This switching is not governed by a single threshold, but by an interaction between stress and retained elasticity, forming a sloped regime boundary. Household-channel leakage dominates at moderate stress levels when sufficient elasticity remains, while market-channel leakage dominates when stress is high or elasticity is depleted.

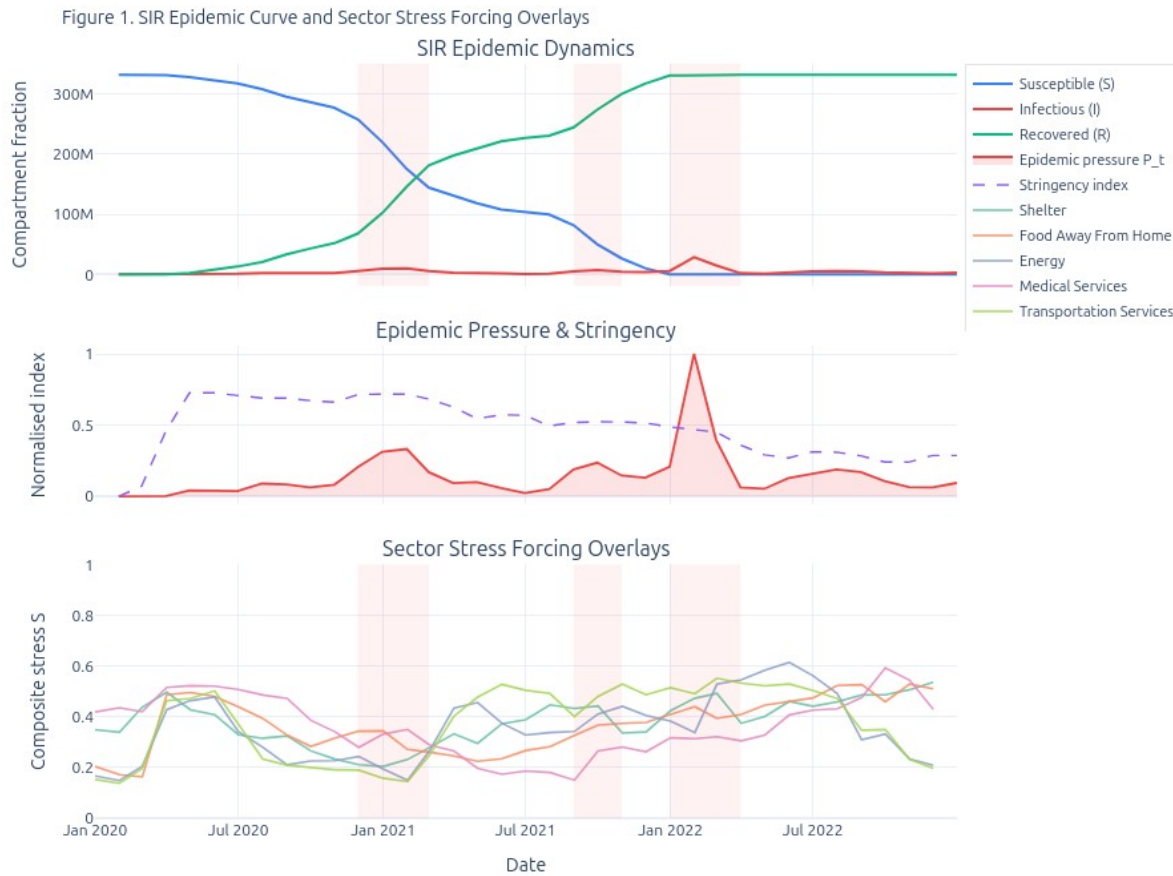


Figure 1: Epidemic forcing and sectoral exposure profiles

Time series of epidemic prevalence with sector-specific forcing overlays. This figure defines the external pressure field entering the system, highlighting differences in magnitude and timing across sectors that drive subsequent stress accumulation.

Figure 1 should be interpreted not as a descriptive time series, but as the external pressure field that drives all subsequent system dynamics. These findings imply that systems may undergo significant internal degradation before this degradation becomes visible through conventional market indicators. In particular, stress can be displaced into household channels during a transition regime, delaying observable structural failure. This suggests that monitoring only market-based signals may systematically underestimate the extent of system strain prior to regime shifts.

The paper proceeds as follows. Section 2 motivates the coupled modeling approach and situates it relative to existing work. Section 3 introduces the SER framework and its coupling to SIR epidemic dynamics. Section 4 presents the formal model, including dynamic connectivity and leakage decomposition. Section 5 describes the data and simulation setup. Section 6 presents baseline results, followed by distributional analysis of household-channel leakage in Section 7. Section 8 reports results from a parameter sweep demonstrating the robustness of competing exit channels, and Section 9 derives the regime boundary governing channel dominance. Section 10 discusses implications, limitations, and directions for further work.

2. Background and Motivation

2.1 Epidemic shocks as system-wide constraint events

Epidemic events are typically analyzed through two separate lenses: public health dynamics and economic impact. In public health, models such as the susceptible–infected–recovered (SIR) framework capture the temporal evolution of disease spread. In economics, the effects of epidemics are often treated as exogenous shocks to demand, supply, or labor availability, reflected in output contraction, price changes, or employment shifts.

However, this separation obscures an important feature of large-scale disruptions: epidemic shocks act as system-wide constraint events. They simultaneously alter participation, mobility, production capacity, and behavioral responses across multiple sectors. These effects do not enter the economy as a single impulse, but as a time-varying pressure field that interacts with existing structural dependencies.

As a result, the economic impact of an epidemic cannot be fully characterized by immediate changes in output or prices. Instead, it must be understood as a process in which burden accumulates, is partially absorbed, and is redistributed across interconnected sectors before ultimately being released.

2.2 Limitations of standard economic representations

Conventional economic approaches often assume that shocks are either:

- absorbed through price adjustments,
- transmitted through input-output relationships, or
- realized as observable losses in output or employment.

While these mechanisms are important, they implicitly assume that stress is either:

1. fully absorbed within the system, or
2. immediately observable through formal market channels.

This leaves a gap in representation. In practice, systems can remain operational even while absorbing substantial burden outside observable market activity. For example:

- households may defer consumption,
- firms may temporarily reduce throughput without immediate failure,
- participation may decline without corresponding price signals.

These behaviors do not necessarily appear in standard metrics, yet they represent real system strain. Without an explicit representation of how unresolved burden exits the system, such dynamics remain difficult to detect and interpret.

2.3 The SER framework as a state-based approach

The Stress–Elasticity–Reaction (SER) framework addresses this gap by modeling economic systems in terms of state variables rather than equilibrium conditions. In this formulation:

- **Stress** (S) represents accumulated burden or unmet demand on a sector,
- **Elasticity** (E) represents the sector’s capacity to absorb, adapt, or buffer that burden,
- **Reaction** (R) represents observable system response, including measurable instability or output change.

This approach allows for:

- delayed responses,
- partial absorption of shocks,
- and non-linear transitions when capacity is exceeded.

Unlike equilibrium-based models, the SER framework explicitly accommodates the possibility that stress may persist without immediate resolution, and that system behavior depends on the evolving relationship between burden and capacity.

2.4 Coupling SER with SIR dynamics

While SER provides a representation of internal system dynamics, it does not specify the origin or time profile of external shocks. Epidemic processes, by contrast, provide a structured and empirically grounded source of time-varying pressure.

By coupling SIR epidemic dynamics to the SER framework, we obtain a model in which:

- epidemic prevalence generates sector-specific forcing,
- this forcing accumulates as stress within sectors,
- and system response depends on evolving elasticity and network interactions.

Formally, epidemic prevalence (I_t^{sir}) is mapped into sector-level pressure through exposure weights and lags:

$$P_{i,t}^{covid} = w_i \cdot I_{t-l_i}^{sir}$$

This mapping allows different sectors to experience the epidemic shock with different magnitudes and timing, reflecting heterogeneity in exposure and response.

2.5 The missing component: exit channels for leaked stress

Even with coupled SER-SIR dynamics, a critical question remains:

What happens to stress that exceeds a sector’s absorptive capacity?

Standard formulations implicitly assume that such stress appears as observable failure. However, empirical observation suggests that this is not always the case. Systems often continue functioning despite significant underlying strain, implying that burden is being rerouted rather than immediately realized.

To address this, we introduce a decomposition of leaked stress into two exit channels:

- **Market-channel leakage** ((L_{market})), representing stress that propagates through observable exchange mechanisms and institutional structures,
- **Household-channel leakage** ($(L_{\text{household}})$), representing stress that propagates through human, local, and temporal mechanisms outside standard measurement.

This decomposition does not assume direct observability of household-channel dynamics. Instead, it treats them as an inferred residual, identified through divergence between expected system performance and observed activity.

2.6 Motivation for regime analysis

The introduction of multiple exit channels raises a further question:

Under what conditions does the system route stress through one channel rather than the other?

If the choice of exit channel depends systematically on system state, then the system may exhibit distinct regimes of behavior. In particular, it is plausible that:

- when sufficient elasticity remains, stress is displaced into household channels,
- when elasticity is depleted, stress is forced into market-visible failure.

This suggests the existence of a regime boundary in the joint space of stress and elasticity.

The remainder of the paper investigates this hypothesis by:

- simulating the coupled SER-SIR system,
- analyzing the statistical properties of leakage channels,
- and performing a parameter sweep to determine whether exit-channel dominance exhibits robust, structured switching behavior.

Figure 2. Conceptual Layering of the SER–SIR Model

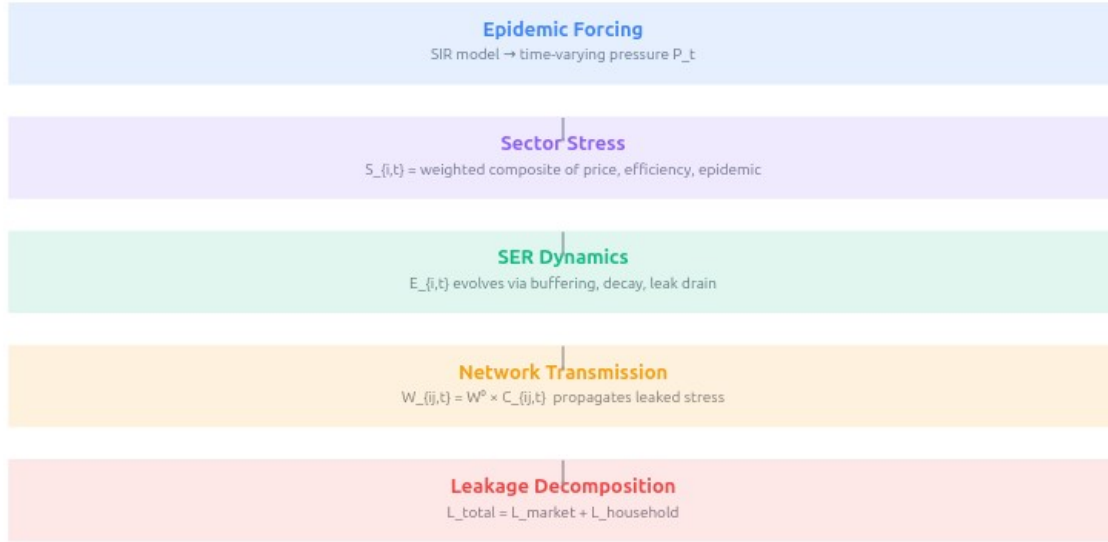


Figure 2: Conceptual layering of the model. SIR epidemic dynamics generate time-varying external pressure, which is mapped into sectoral stress within the SER framework. Stress interacts with elasticity and network structure, producing leakage that exits through market and household channels.

3. Conceptual Framework and Formal Specification

This section presents the formal structure of the coupled epidemic–economic system. The model combines:

- an epidemic forcing layer (SIR),
- a sectoral state model (SER),
- a dynamic network transmission layer,
- and a leakage decomposition that defines exit channels for unresolved stress.

The goal is not to construct a fully structural macroeconomic model, but to specify a state-based dynamical system in which stress accumulation, absorption, transmission, and release can be explicitly represented.

3.1 SER state variables

For each sector (i) and time (t), we define three primary state variables:

$S_{i,t}$ (stress)

$E_{i,t}$ (*elasticity*)

$R_{i,t}$ (*reaction*)

Interpretation

- **Stress** ($S_{i,t}$) represents accumulated burden, including unmet demand, operational constraints, or external pressure that has not yet been resolved.
- **Elasticity** ($E_{i,t}$) represents the capacity of the sector to absorb, adapt to, or buffer incoming stress.
- **Reaction** ($R_{i,t}$) represents observable system response, including output change, instability, or price/throughput effects.

This formulation allows stress to persist and accumulate even when observable reaction remains limited, capturing the possibility of delayed or masked system responses.

3.2 Epidemic forcing (SIR layer)

We represent epidemic dynamics using a standard susceptible–infected–recovered (SIR) process:

$$\left[\frac{dS^{sir}}{dt} = -\beta S^{sir} I^{sir} \right]$$

$$\left[\frac{dI^{sir}}{dt} = \beta S^{sir} I^{sir} - \gamma I^{sir} \right]$$

$$\left[\frac{dR^{sir}}{dt} = \gamma I^{sir} \right]$$

or, in discrete form:

$$[I_{t+1}^{sir} = I_t^{sir} + \beta S_t^{sir} I_t^{sir} - \gamma I_t^{sir}]$$

Mapping to economic pressure

Epidemic prevalence is mapped into sector-specific pressure:

$$[P_{i,t}^{covid} = w_i \cdot I_{t-l_i}^{sir}]$$

where:

- (w_i) is a sector-specific exposure weight,
- (l_i) is a sector-specific lag.

This allows epidemic pressure to enter different sectors at different magnitudes and times.

3.3 Stress dynamics

Sectoral stress evolves as a function of prior stress, external forcing, and transmitted pressure:

$$[S_{i,t+1} = (1 - \delta)S_{i,t} + \alpha B_{i,t} + \chi P^{covid}_{i,t} + \lambda P^{net}_{i,t}]$$

where:

- (δ) : stress decay or resolution rate
- $(B_{i,t})$: baseline or endogenous burden
- $(P^{covid}_{i,t})$: epidemic forcing
- $(P^{net}_{i,t})$: network-transmitted stress
- (α, χ, λ) : scaling parameters

Interpretation

Stress accumulates through both direct epidemic forcing and indirect transmission from other sectors, while partially decaying over time.

3.4 Elasticity dynamics

Elasticity evolves as a stock variable subject to both reinforcement and depletion:

$$[E_{i,t+1} = E_{i,t} + \alpha_E H_{i,t} - \beta_E S_{i,t} - \eta_E L_{i,t}]$$

where:

- $(H_{i,t})$: reinforcement or recovery input (e.g., adaptation, support)
- $(S_{i,t})$: stress-induced erosion
- $(L_{i,t})$: leakage-induced erosion
- $(\alpha_E, \beta_E, \eta_E)$: parameters

Interpretation

Elasticity is a **buffer stock**:

- it is depleted by persistent stress and leakage,
- and replenished by adaptive or supportive processes.

Empirically, this state variable emerges as the primary determinant of exit-channel dominance.

3.5 Reaction dynamics

Observable system response is modeled as a function of leakage and elasticity:

$$[R_{i,t} = aL_{i,t} + a_{qL}^{queue} i_{i,t} + a_{aL}^{accept} i_{i,t} + a_{\epsilon} L_{i,t}^{loss} - bE_{i,t} + \gamma R_{i,t-1} + \epsilon_{i,t}]$$

Interpretation

Reaction reflects:

- the manifestation of leaked stress,
- modulated by remaining elasticity,
- with persistence through time.

3.6 Field mismatch representation

We define a mismatch field:

$$[\Phi_{i,t} = R_{i,t} - S_{i,t}]$$

Interpretation

$(\Phi_{i,t})$ captures the alignment between:

- underlying stress, and
- observable system response.

Large mismatches indicate delayed or incomplete system adjustment, and can be used to modulate transmission between sectors.

3.7 Dynamic connectivity

Structural prior

$$[W_{ij}^0 = \text{baseline coupling between sectors } i \text{ and } j]$$

Evolving connectivity

$$[C_{ij,t} = (1 - \rho_c) C_{ij,t-1} + \rho_c G_{ij,t}]$$

$$[W_{ij,t} = W_{ij}^0 \cdot C_{ij,t}]$$

where $(G_{ij,t})$ is a function of joint sector state.

A representative realization:

$$[G_{ij,t} = \sqrt{E_{i,t} E_{j,t}} \cdot \sqrt{A_{i,t} A_{j,t}} \cdot \left(1 - \frac{|\Phi_{i,t} - \Phi_{j,t}|}{4}\right) \cdot L_{ij,t}^{avg}]$$

Interpretation

Connectivity becomes an **evolving transmission state**, reflecting:

- joint capacity (elasticity),
- realized activity,

- mismatch in system response,
- and leakage pressure.

This replaces static or diagnostic connectivity with a state variable that captures contraction and recovery of transmission pathways.

3.8 Network transmission of stress

Network-transmitted stress is defined as:

$$[P^{net}_{i,t} = \sum_j W_{ij,t} \cdot L_{j,t}]$$

Interpretation

Stress propagates across sectors through dynamically weighted connections, allowing for:

- amplification,
- attenuation,
- and spatial/structural diffusion of burden.

3.9 Leakage decomposition

Total leakage is decomposed into two exit channels:

$$[L_{i,t} = L_{market,i,t} + L_{household,i,t}]$$

3.9.1 Market-channel leakage

$$[L_{market,i,t} = \max(0, S_{i,t} - S_{i,t}^{absorbed})]$$

This represents stress that exits into:

- observable exchange systems,
- institutional channels,
- and measurable economic disruption.

3.9.2 Household-channel leakage

Household leakage is inferred as a residual:

$$[A_{i,t}^{expected} = \text{clip}((1 - S_{i,t}) \cdot (0.6 + 0.25 E_{i,t}^{norm} + 0.15 W_{i,t}^{in,norm}), 0, 1)]$$

$$[L_{household,i,t} = \text{clip}(A_{i,t}^{expected} - A_{i,t}^{observed}, 0, 1)]$$

Interpretation

- $((L_{household}))$ represents stress exiting through:

- deferral,
 - substitution,
 - reduced participation,
 - and other human-domain mechanisms.
- It is **not directly observed**, but inferred from divergence between:
 - expected system performance given capacity,
 - and realized activity.

Empirically, this channel exhibits:

- non-Gaussian distributions,
- zero-inflation,
- and episodic activation consistent with a boundary exit process.

3.10 System summary

The full system can be summarized as:

1. Epidemic dynamics generate time-varying pressure.
2. Pressure enters sectors as stress.
3. Stress interacts with elasticity and network structure.
4. Unresolved stress leaks out through two competing channels:
 - market (observable),
 - household (latent).
5. The dominance of these channels depends on the joint stress–elasticity state.

Figure 3. Full System Diagram — Coupled SER–SIR Architecture

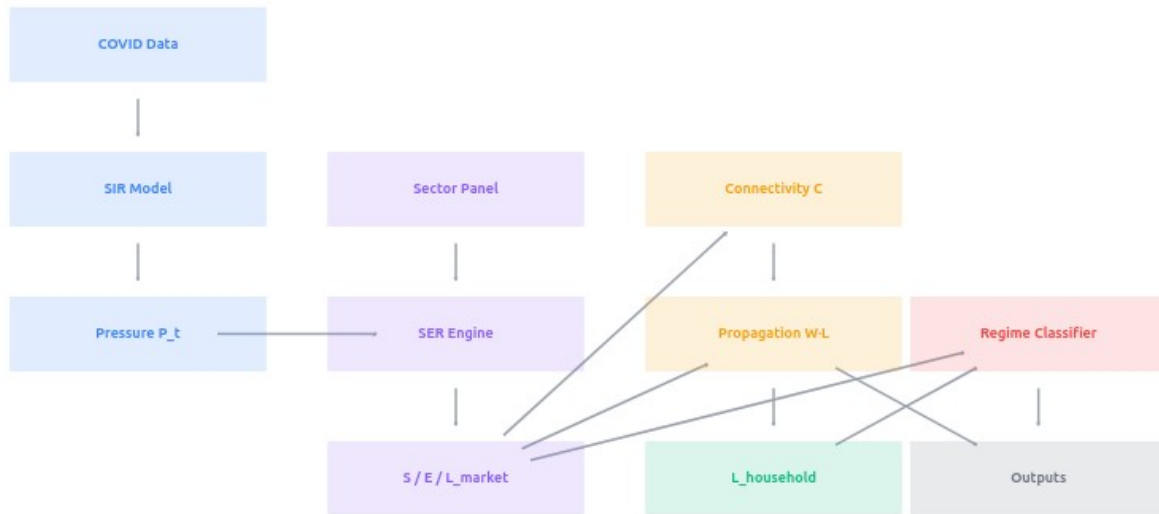


Figure 3: Full System Diagram. Coupled SER–SIR system showing epidemic forcing, sectoral stress and elasticity dynamics, network transmission, dynamic connectivity, and leakage decomposition into market and household exit channels.

4. Data and Simulation Setup

This section describes the data sources, sector construction, and simulation procedures used to implement the coupled SER–SIR system. The goal is not to reconstruct a full national accounting framework, but to assemble a consistent, reproducible dataset that allows the dynamics of stress, elasticity, transmission, and leakage to be observed under realistic conditions.

4.1 Epidemic data and SIR construction

Epidemic forcing is derived from publicly available COVID-19 data (e.g., case counts or estimated active infections). The data are used to construct a time series for epidemic prevalence, which is then mapped into an SIR process.

Implementation steps

1. Input data

- Daily or weekly COVID-19 case counts (e.g., from OWID or CDC aggregates)

2. Smoothing and normalization

- Rolling averages applied to reduce reporting noise

- Series normalized to represent a share of population or relative intensity

3. SIR parameterization

- Infection rate (β)
- Recovery rate (γ)

4. Discrete update

$$[I_{t+1}^{sir} = I_t^{sir} + \beta S_t^{sir} I_t^{sir} - \gamma I_t^{sir}]$$

The resulting (I_t^{sir}) serves as the epidemic pressure signal used in the economic model.

4.2 Sector construction

The economic system is represented as a set of aggregated sectors selected to capture heterogeneous exposure to epidemic and economic conditions.

Representative sector set

- Food at home
- Food away from home (restaurants, hospitality)
- Medical services
- Household goods
- Energy
- Electricity
- Transportation services
- Shelter (housing)
- Gasoline
- Apparel / general merchandise

These sectors are chosen to reflect:

- varying exposure to mobility and labor shocks,
- differences in demand elasticity,
- differing roles in household vs institutional consumption.

4.3 Economic data and proxies

Sector-level state variables are not directly observed. Instead, they are constructed from publicly available proxies.

Data sources

- **FRED (Federal Reserve Economic Data)**
 - CPI/PPI series
 - sectoral price indices
 - production/utilization indicators
- **Opportunity Insights (OI) Economic Tracker (if available)**
 - consumer spending proxies
 - sectoral activity indicators

Variable construction

- **Observed activity** ($A_{i,t}^{observed}$)
 - constructed from spending or proxy activity series
 - normalized to $[0, 1]$
- **Baseline burden** ($B_{i,t}$)
 - derived from deviations in price, output, or utilization
- **Elasticity proxies**
 - inferred from combinations of:
 - stability of activity,
 - responsiveness to stress,
 - recovery behavior

Important constraint

To avoid circularity:

$$(A_{i,t}^{observed}) \text{ is constructed independently of } (1 - S_{i,t})$$

This ensures that household leakage is not mechanically derived from stress.

4.4 Sectoral epidemic exposure

Epidemic forcing is mapped into sectors using exposure weights and lags:

$$[P_{i,t}^{covid} = w_i \cdot I^{sir}_t - l_i]$$

Where:

- (w_i) reflects sector sensitivity (e.g., restaurants high, utilities low)

- (l_i) reflects delay between infection prevalence and sector impact

Example mapping logic

- High-contact sectors (food away, transport): high (w_i) , short lag
- Essential services (utilities, medical): moderate (w_i) , moderate lag
- Household goods: lower (w_i) , longer lag

This mapping introduces heterogeneity in how epidemic pressure enters the economic system.

4.5 Network structure and initialization

Structural prior

A baseline connectivity matrix (W_{ij}^0) is defined using:

- expert-informed sector relationships, or
- simplified input-output structure

Initialization

- $(C_{ij,0}=1)$ for all edges
 - initial stress $(S_{i,0})$ set to baseline
 - initial elasticity $((E_{i,0}))$ drawn from sector-specific ranges
-

4.6 Simulation procedure

The system is simulated over discrete time steps $(t=1, \dots, T)$, typically covering the duration of the epidemic period.

Per-step update sequence

For each time step:

1. **Update epidemic state**
 - compute (I_t^{sir})
2. **Compute sectoral epidemic forcing**
 - $(P_{i,t}^{covid})$
3. **Update stress**

- using stress dynamics equation

4. Update elasticity

- using elasticity dynamics equation

5. Compute leakage

- $(L_{market,i,t})$
- $(L_{household,i,t})$

6. Update connectivity

- $(C_{ij,t}), (W_{ij,t})$

7. Compute network transmission

- $(P_{i,t}^{net})$

8. Update reaction

- $(R_{i,t})$

This ordering ensures that:

- leakage affects future elasticity,
- connectivity reflects current state,
- and transmission carries forward dynamically.

4.7 Activity Structure Fusion and Blob-Based Topology Inference

To capture dynamic changes in economic structure, sectoral connectivity is informed by a fusion of:

- **Benchmark Input-Output structure (BEA IO tables)**, providing baseline inter-sector relationships, and
- **Observed activity data (Opportunity Insights Economic Tracker)**, providing time-varying sectoral intensity.

Rather than treating connectivity as fixed or purely parametric, this approach allows for **inference of evolving structure** through the joint behavior of activity magnitude and co-movement.

Let:

$$A_{i,t} \in [0, 1]$$

represent normalized sectoral activity. Define a similarity kernel:

$$K_{ij,t} = \exp\left(-\frac{|A_{i,t} - A_{j,t}|}{\sigma}\right)$$

and combine with structural prior:

$$W_{ij,t} = W_{ij}^0 \cdot K_{ij,t}$$

This formulation produces **dynamic clustering behavior**, where sectors with similar activity levels and strong structural ties form **coherent clusters (“blobs”)**.

4.8 Diagnostics and validation checks

Several diagnostics are performed to ensure the model behaves as intended:

Connectivity check

- Verify that $(C_{ij,t})$ evolves over time (non-zero variance)
- Confirm removal of flat-connectivity artifact

Leakage behavior

- Verify that $(L_{household})$ is not trivially derived
- Check independence from stress proxies

Stability checks

- confirm bounded state variables
- ensure no runaway divergence without parameter cause

4.9 Parameter sweep design

To test robustness, a Latin hypercube parameter sweep is conducted.

Sweep configuration

- **200 runs**
- **8 parameters varied:**
 - stress decay (δ)
 - leakage sensitivity (β_L)
 - stress scaling (α)
 - epidemic coupling (χ)
 - epidemic weight (w_{covid})
 - connectivity persistence (ρ_c)
 - elasticity scaling (k_E)
 - initial elasticity (E_{init})

Each run produces sector-level time series, yielding approximately 2,000 sector-run observations.

4.10 Output variables

For each sector and run, the following are recorded:

- $(S_{i,t}), (E_{i,t}), (R_{i,t})$
- $(L_{market,i,t})$
- $(L_{household,i,t})$
- $(C_{ij,t})$
- derived statistics:
 - means
 - peaks
 - correlations
 - regime labels

Figure 4. Data and Simulation Pipeline

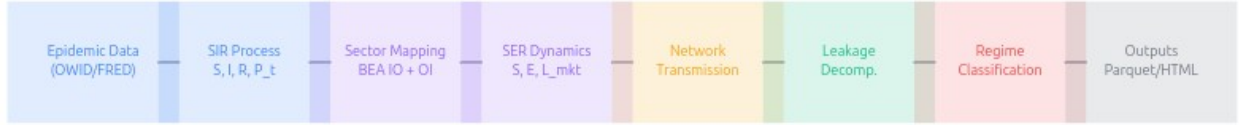


Figure 4: Data and simulation pipeline. Flow diagram from epidemic data → SIR process → sector mapping → SER dynamics → network transmission → leakage decomposition → outputs.

5. Baseline Results: Coupled SER–SIR Dynamics (Draft)

This section presents baseline simulation results for the coupled SER–SIR system. The goal is to illustrate how epidemic forcing propagates through sectoral stress, elasticity, network transmission, and leakage before introducing distributional tests and parameter-sweep results.

5.1 Epidemic forcing and sectoral stress

Epidemic prevalence ($I^{sir}t$) generates a time-varying pressure signal that is mapped into sector-specific forcing ($P^{covid}i,t$). Due to heterogeneous exposure weights and lags, sectors do not respond uniformly.

Observations

- High-contact sectors (e.g., food away from home, transportation) exhibit earlier and larger stress increases.
- Essential or buffered sectors (e.g., utilities, shelter) exhibit delayed and attenuated responses.
- Stress trajectories are asynchronous across sectors, producing dispersion rather than uniform movement.

Interpretation

Epidemic pressure enters the economic system as a **distributed, time-shifted forcing field**, rather than a single synchronized shock.

5.2 Elasticity erosion and buffering

Elasticity ($E_{i,t}$) evolves as a stock that is gradually depleted under sustained stress.

Observations

- Sectors with sustained high stress show **progressive elasticity erosion**.
- Temporary stress spikes do not immediately collapse elasticity; depletion requires persistence.
- Recovery (where present) occurs slowly and unevenly across sectors.

Interpretation

Elasticity acts as a **buffer stock**:

- it delays the manifestation of stress,
- and determines how long sectors can absorb pressure before leakage increases.

This aligns with later results showing elasticity as the primary determinant of exit-channel dominance.

5.3 Network transmission and dynamic connectivity

The introduction of dynamic connectivity ($C_{ij,t}$) produces evolving transmission pathways.

Observations

- All edges exhibit **non-zero variance in connectivity over time**, confirming removal of the flat-connectivity artifact.
- Connectivity tends to **decline under high stress and leakage**, reflecting reduced transmission capacity.
- Recovery of connectivity is gradual and uneven, reflecting dependence on joint sector state.

Interpretation

Connectivity behaves as a **state variable** rather than a fixed property:

- it contracts under stress,
- and reopens as conditions improve.

This allows the model to represent **changing transmission capacity**, rather than assuming static interdependence.

5.4 Leakage dynamics over time

Total leakage ($L_{i,t}$) decomposes into market and household channels.

Observations

- In early phases, leakage is low and primarily absorbed within sectors.
- As stress accumulates, leakage increases, but its **channel composition varies over time**.
- Periods of moderate stress often show **higher household-channel leakage**, while later stages shift toward market-channel dominance.

Interpretation

Leakage is not a single outcome but a **routing process**, with stress exiting through different channels depending on system state.

5.5 Reaction and observable outcomes

Reaction ($R_{i,t}$) reflects the visible manifestation of system dynamics.

Observations

- Reaction lags stress in many sectors, indicating delayed adjustment.
- Peaks in reaction often occur after significant elasticity erosion.
- Reaction is more closely aligned with **market-channel leakage** than with total stress.

Interpretation

Observable outcomes are **downstream of leakage and elasticity**, not direct reflections of stress alone. This helps explain why systems can exhibit internal strain before visible instability emerges.

5.6 Integrated system behavior

Combining these components yields a coherent dynamic:

1. Epidemic forcing raises sector-specific stress.
2. Elasticity buffers stress initially but erodes over time.

3. Connectivity contracts as joint sector conditions deteriorate.
4. Unresolved stress exits through leakage channels.
5. Observable reaction emerges as leakage increases and elasticity declines.

Key insight

Even in the baseline run, prior to parameter sweeps:

The system exhibits **phased behavior**, with distinct periods of buffering, transition, and visible breakdown.

Figure 5. Sectoral Stress Trajectories Under Epidemic Forcing



Figure 5: Sectoral stress trajectories. Time series of $(S_{i,t})$ for representative sectors under epidemic forcing.

Figure 6. Elasticity Trajectories — Depletion and Partial Recovery



Figure 6: Elasticity trajectories. Evolution of $(E_{i,t})$ showing gradual depletion and partial recovery.

Figure 7. Connectivity Evolution — Selected Edge Histories $C_{ij,t}$

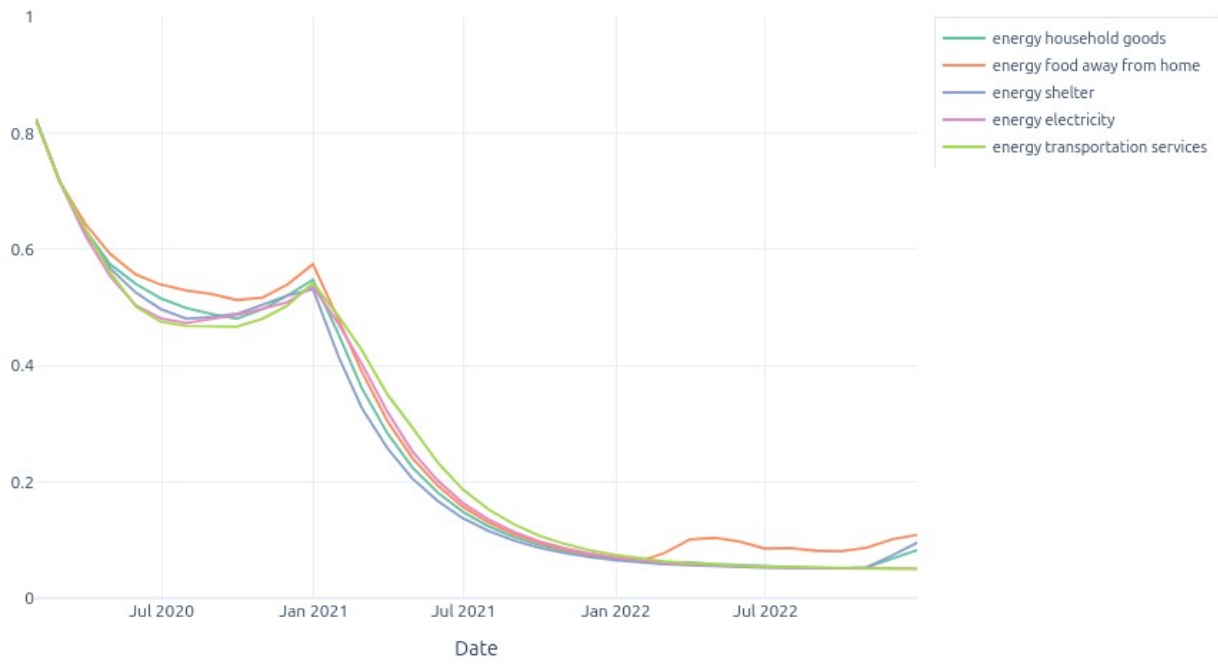


Figure 7: Connectivity evolution. Selected $(C_{ij,t})$ edge histories illustrating contraction and recovery of transmission capacity.

Figure 8. Leakage Decomposition — L_{market} and $L_{\text{household}}$ Over Time

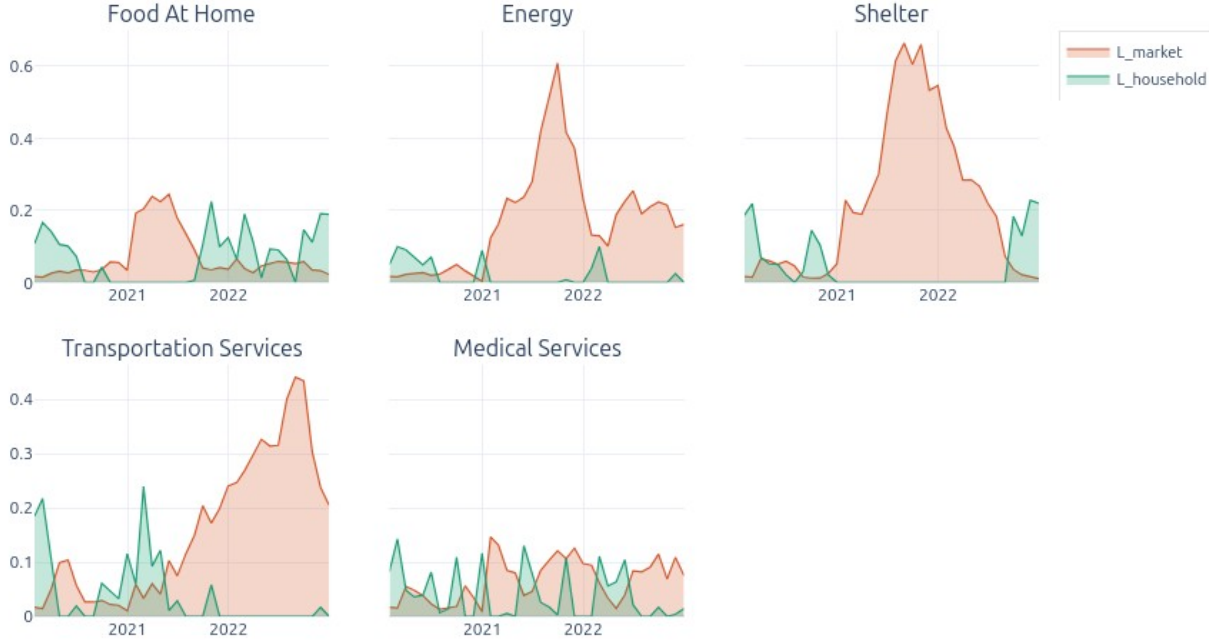


Figure 8: Leakage decomposition over time

Stacked time series of (L_{market}) and $(L_{\text{household}})$ for selected sectors.

6. Statistical Character of Household-Channel Leakage (Draft)

This section examines the statistical properties of household-channel leakage $(L_{\text{household},i,t})$. The goal is to determine whether it behaves as a conventional bounded state variable or as a distinct process associated with boundary crossing and episodic activation.

6.1 Analytical objective

Household-channel leakage is defined as an inferred residual between expected system performance and observed activity. As such, it is not directly observed and does not have an a deductive distributional form.

The key question is:

Does $(L_{\text{household}})$ behave as a continuous, in-system stochastic variable, or as an intermittently activated exit process?

This distinction is important because it determines whether the variable should be interpreted as:

- part of the internal system state, or
- a boundary mechanism through which stress exits the modeled domain.

6.2 Distributional properties

We analyze the distribution of $(L_{household})$ across sectors and time using standard statistical diagnostics.

Summary statistics

Across sectors:

- Mean values are low relative to total stress.
- Standard deviations vary across sectors but remain bounded.
- Distributions exhibit:
 - **positive skewness** (up to ~ 2.6),
 - **excess kurtosis** (up to ~ 7.2).

Normality tests

Standard tests (e.g., Shapiro–Wilk, Jarque–Bera, Anderson–Darling) consistently reject normality across sectors.

Interpretation

The combination of skewness, kurtosis, and rejection of normality indicates that:

$(L_{household})$ is not well-described by a Gaussian or near-Gaussian process.

6.3 Zero-inflation and activation behavior

A defining feature of $(L_{household})$ is its distribution near zero.

Observations

- A substantial fraction of observations lie near zero:
 - approximately **36%–75%** depending on sector.
- Non-zero values appear in **episodic bursts**, typically comprising:
 - **5%–17%** of observations.
- Transitions from near-zero to positive values often occur **abruptly within a single time step**.

Interpretation

This pattern is consistent with:

- a **zero-inflated process**, and
- an **intermittently activated channel**.

In contrast to continuous variables, $(L_{household})$ appears to represent a mechanism that is:

- inactive most of the time,
- but activates under specific conditions.

6.4 Temporal behavior and persistence

To assess whether $(L_{household})$ exhibits stable statistical behavior over time, we analyze rolling and expanding window statistics.

Observations

- Rolling means and variances do not converge to stable values.
- Skewness and kurtosis vary significantly over time.
- Periods of activation are followed by rapid reversion toward zero.

Interpretation

The lack of convergence suggests that:

$(L_{household})$ does not exhibit stationary statistical behavior consistent with an internal system variable.

Instead, it behaves as a **non-stationary, event-driven process**.

6.5 Mixture and breakpoint characteristics

We test whether the distribution of $(L_{household})$ is better described by a mixture or regime-switching process.

Observations

- In several sectors, two-component mixture models outperform single-distribution fits.
- Change-point detection identifies multiple **structural breaks** aligned with activation periods.
- Threshold-based behavior is evident:
 - activation occurs disproportionately within specific stress ranges.

Interpretation

These results indicate that:

$(L_{household})$ is better characterized as a mixture or regime-switching process than as a single continuous distribution.

6.6 Exit-channel interpretation

Taken together, the distributional evidence supports a specific interpretation:

- $(L_{household})$ is not a smooth function of system state,
- it is not continuously active,
- and it does not behave as a stable stochastic component.

Instead:

$(L_{household})$ behaves as a **boundary exit channel**, which is typically inactive but activates intermittently when system conditions require stress to be routed outside observable market structures.

This interpretation is consistent with its construction as a residual and with its observed episodic activation.

6.7 Implications for modeling

The statistical behavior of $(L_{household})$ implies that:

1. It should not be modeled as a standard continuous variable.
2. Its primary characteristics are:
 - activation probability,
 - episode duration,
 - and activation conditions.
3. Its role is to indicate **when and where stress exits the modeled system through non-market channels**.

This perspective motivates the regime analysis in subsequent sections, where the conditions governing the activation and dominance of leakage channels are examined.

Figure 9. Distribution of L_household — Histograms and KDE

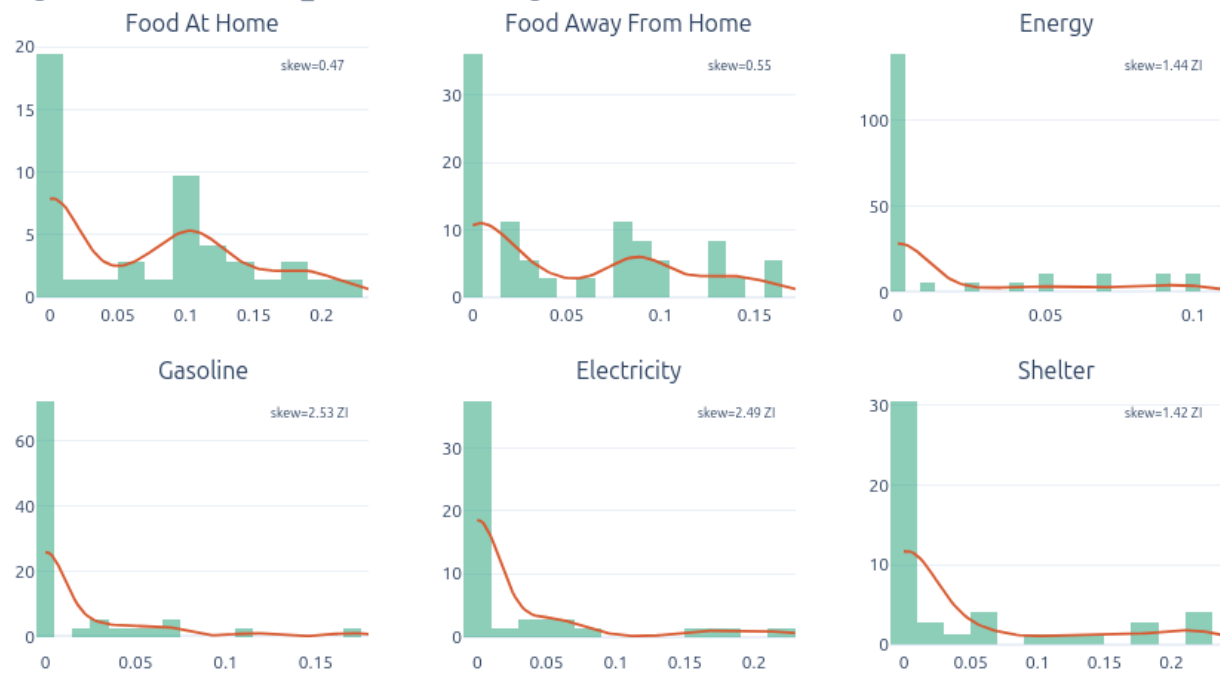


Figure 9: Distribution of household-channel leakage

Histograms and kernel density estimates showing skewness, kurtosis, and zero-inflation across sectors.

Figure 10. Rolling Distribution Diagnostics — L_household (6-period window)



Figure 10: Rolling distribution diagnostics

Time-varying mean, variance, skewness, and kurtosis for representative sectors.

Figure 11. Breakpoint and Activation Analysis — L_household Episodes



Figure 11: Breakpoint and activation analysis. Change-point detection and activation episodes highlighting episodic behavior.

7. Competing Exit Channels: Parameter Sweep Results (Draft)

This section evaluates whether market-channel and household-channel leakage operate independently or as competing exit routes for system stress. To do so, we analyze results from an upstream parameter sweep of the coupled SER–SIR system.

7.1 Objective

The purpose of the parameter sweep is to determine whether the relationship between:

$$[L_{market,i,t} \wedge L_{household,i,t}]$$

is:

- incidental to a specific configuration, or
- a robust structural feature of the system.

In particular, we test the hypothesis that the two channels function as **alternative exit pathways** for the same underlying stress budget.

7.2 Sweep design

A Latin hypercube design is used to sample the parameter space efficiently.

Configuration

- **200 simulation runs**
- **8 parameters varied:**
 - stress decay (δ)
 - leakage sensitivity (β_L)
 - stress scaling (α)
 - epidemic coupling (χ)
 - epidemic weight (w_{covid})
 - connectivity persistence (ρ_c)
 - elasticity scaling (k_E)
 - initial elasticity (E_{init})

Each run generates time series for all sectors, producing approximately **2,000 sector-run observations**.

7.3 Correlation analysis

For each sector in each run, we compute the correlation between:

$$[L_{market,i,t} \wedge L_{household,i,t}]$$

Results

- Correlation is **negative in all 2,000 sector-run observations**.
- Range: approximately **-0.82 to -0.23**.
- No sector in any run exhibits zero or positive correlation.

At the system level:

- **155 out of 200 runs** show correlations below -0.50.
- The remaining runs show weaker but still negative correlations.

Interpretation

The absence of sign changes across the entire parameter space indicates that:

Market-channel and household-channel leakage are not independent processes, but are systematically opposed.

This provides strong evidence that the two channels represent **competing exit routes** for leaked stress.

7.4 Sectoral variation

While the inverse relationship is universal, its strength varies across sectors.

Observations

- Strongest negative correlations:
 - shelter
 - food away from home
- Weaker (but still negative) correlations:
 - gasoline
 - apparel

Interpretation

Differences in correlation strength reflect:

- sector-specific exposure,
- demand characteristics,
- and elasticity dynamics,

but do not alter the fundamental competing-channel structure.

7.5 Robustness to parameter variation

We test whether the inverse relationship is sensitive to individual parameters.

Findings

The relationship remains robust across variation in:

- leakage sensitivity (β_L)
- connectivity persistence (ρ_c)
- elasticity scaling (k_E)
- initial elasticity (E_{init})

These parameters do not significantly alter:

- the sign of correlation,
- or the qualitative structure of channel interaction.

Interpretation

The competing-channel relationship is not driven by:

- specific calibration choices,
- or particular dynamic assumptions,

but appears to be a **structural property of the modeled system** within the explored domain.

7.6 Nonlinear and threshold-based behavior

Despite the consistent inverse relationship, the interaction between channels is not linear.

Observations

- Channel dominance shifts abruptly rather than smoothly.
- Heatmaps reveal **sharp transitions across parameter space**.
- The transition is concentrated within intermediate parameter regions.

Stress-based analysis

When runs are grouped by mean stress levels:

- **Low stress (Q1):**
 - ~80% household-dominant
 - 0% market-dominant
- **High stress (Q4):**
 - ~10% household-dominant
 - ~44% market-dominant
- Transition occurs between:
[$S \approx 0.26 - 0.28$]

Interpretation

The relationship between channels is best described as:

threshold-based competition, rather than continuous trade-off.

7.7 Domain of channel dominance

The sweep identifies distinct regions in parameter space where each channel dominates.

Household-dominant regime

Characterized by:

- **Low-to-moderate stress:**
[$S_{mean} < 0.24$]
- **High elasticity:**
[$E_{mean} > 0.60$]
- **Elevated epidemic pressure with slow decay**
[$w_{covid} > 0.22, quad \delta < 0.07$]

Interpretation

In this regime:

- sectors retain sufficient capacity,
- stress is absorbed and rerouted,
- and leakage occurs primarily through household channels.

Market-dominant regime

Characterized by:

- **High stress:**
[$S_{mean} > 0.29$]
- **Low elasticity:**
[$E_{mean} < 0.35$]
- **Rapid elasticity decay and weak clearance**

Interpretation

In this regime:

- buffering capacity is exhausted,
- stress cannot be rerouted,
- and leakage appears as observable market disruption.

7.8 Role of elasticity as a control variable

Across all analyses, elasticity emerges as the dominant determinant of channel dominance.

Evidence

- Largest difference between regimes occurs in:
[$E_{mean} \approx -0.35(difference)$]
- Parameter variations influence outcomes primarily through their effect on elasticity.

Interpretation

Elasticity functions as a **buffer stock**:

- when sufficient, it enables household-channel routing,
- when depleted, it forces stress into market channels.

7.9 Summary of findings

The parameter sweep establishes that:

1. $(L_{\text{market}}) \wedge (L_{\text{household}})$ are **consistently negatively related** across the explored parameter space.
2. The relationship is **robust** to parameter variation.
3. Channel dominance exhibits **nonlinear, threshold-based switching**.
4. Elasticity is the **primary state variable** governing exit-channel selection.

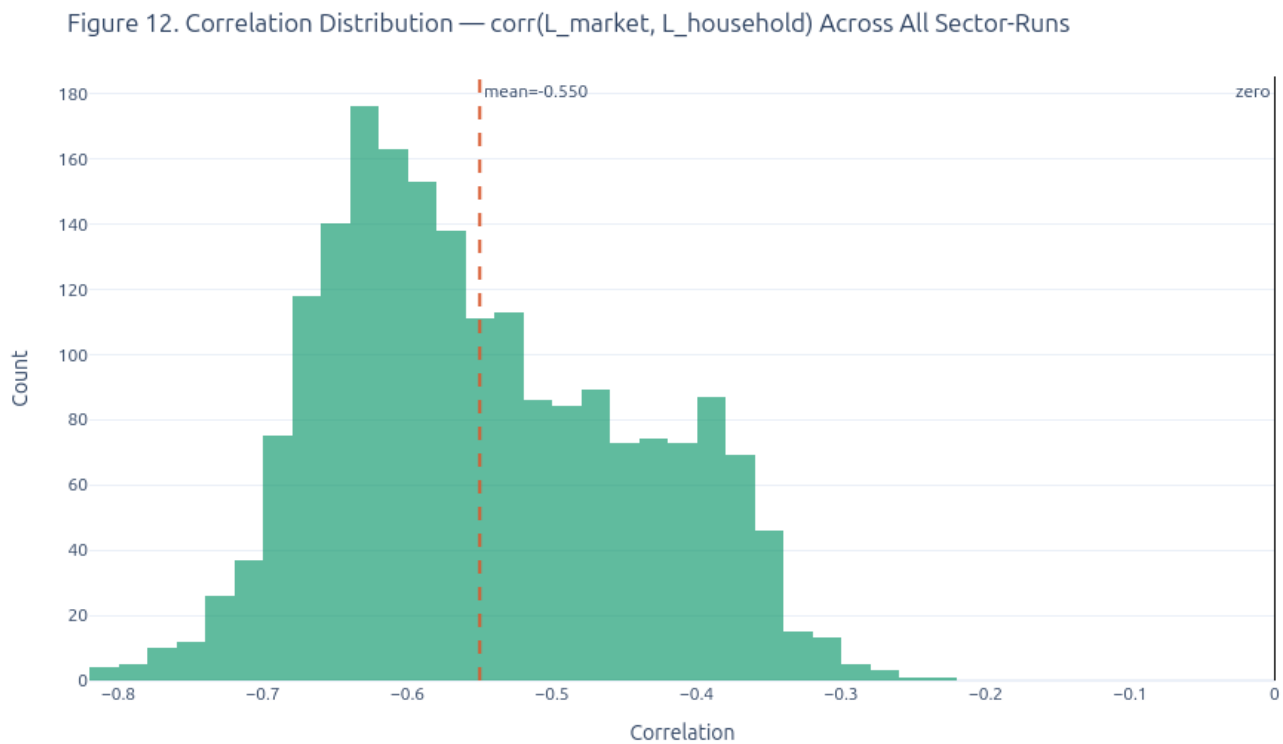


Figure 12: Correlation distribution across runs. Histogram of correlations between market and household leakage across all sector-run observations.

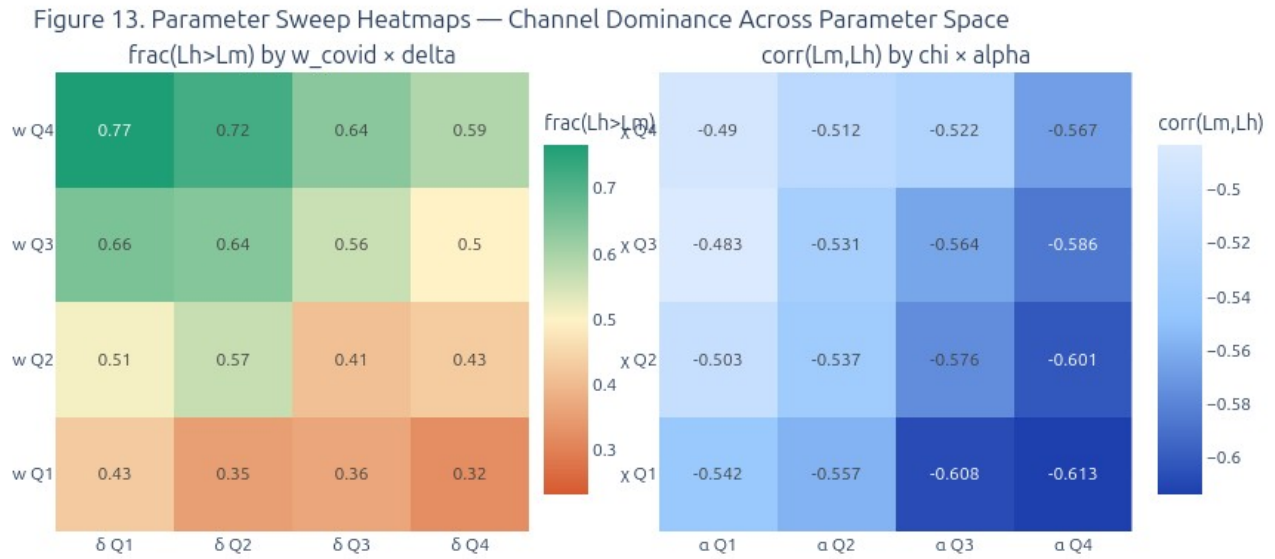


Figure 13: Parameter sweep heatmaps

Regions of household-dominant and market-dominant leakage across selected parameter dimensions.



Figure 14: Stress-based regime distribution. Channel dominance as a function of mean stress quantiles.

8. Regime Boundary and Classifier Results (Draft)

This section formalizes the conditions under which the system switches between household-channel and market-channel leakage. Building on the parameter sweep results, we estimate an explicit **regime boundary** in stress–elasticity space and derive an interpretable decision rule for exit-channel dominance.

8.1 Objective

The parameter sweep establishes that leakage channels are competing and that their dominance is threshold-based. The next step is to determine:

What combination of system states (S, E) determines which channel is active?

We approach this by fitting classifiers that map system state variables to regime labels.

8.2 Regime labeling

Each sector-run observation is labeled based on mean leakage over time:

- **Household-dominant regime:**
 $[\bar{L}_{household} > \bar{L}_{market}]$
- **Market-dominant regime:**
 $[\bar{L}_{market} \geq \bar{L}_{household}]$

This binary labeling enables supervised classification.

8.3 Classifier performance

We evaluate multiple classifiers, including:

- decision trees,
- logistic regression,
- random forests.

Results

- Cross-validated accuracy: **88%–94%**
- All classifiers converge on similar decision boundaries.

Key finding

A shallow decision tree ($depth \leq 4$) achieves comparable accuracy using only:

$[S_{mean}, E_{mean}]$

Interpretation

Exit-channel dominance is governed primarily by the joint stress–elasticity state, with minimal dependence on other parameters once these variables are known.

8.4 Interpretable decision rule

The decision tree yields the following rule:

```
IF  $E_{mean} \leq 0.48$  → MARKET DOMINANT
ELIF  $S_{mean} > 0.31$  → MARKET DOMINANT
ELIF  $S_{mean} \leq 0.27$  → HOUSEHOLD DOMINANT
ELIF  $S_{mean} \in (0.27, 0.31)$  AND  $E > 0.61$  → HOUSEHOLD DOMINANT
ELIF  $S_{mean} \in (0.27, 0.31)$  AND  $E \leq 0.61$  → MARKET DOMINANT
```

Interpretation

This rule reveals a two-stage structure:

1. Elasticity floor condition:

- If $(E_{mean} \leq 0.48)$, the system is market-dominant regardless of stress.
- Elasticity is a **necessary condition** for household-channel routing.

2. Stress overload condition:

- If $(S_{mean} > 0.31)$, the system is market-dominant regardless of elasticity.
- Stress acts as a **sufficient condition** for market-channel activation.

3. Intermediate regime:

- In the range $(S \in (0.27, 0.31))$, dominance depends on whether elasticity exceeds a higher threshold $((E > 0.61))$.

8.5 Continuous boundary estimation

To capture the transition more precisely, we fit a logistic model:

$$[P(\text{household} - \text{dominant}) = \sigma(a + bS_{mean} + cE_{mean})]$$

Results

- Decision boundary slope:

$$[\frac{d\hat{E}}{d\hat{S}} \approx 2.54]$$

- Centroid:

$$[S^{crit} \approx 0.265, E^{crit} \approx 0.521]$$

Interpretation

The boundary is **sloped**, indicating that:

The system does not switch at fixed thresholds of stress or elasticity alone, but through their interaction.

Higher stress requires higher elasticity to maintain household-channel routing.

8.6 Switching band

Rather than a sharp boundary, the transition occurs across a finite-width band.

Definition

- Logistic probability band:
 $[P \in [0.4, 0.6]]$

Observed region

$[S \in [0.22, 0.31], E \in [0.28, 0.76]]$

Interpretation

This region represents:

- instability or sensitivity zone**, where:
 - small perturbations change the dominant exit channel,
 - both channels are viable,
 - system behavior is least predictable.

8.7 Feature importance

Permutation importance provides a robust measure of variable influence.

Results

Feature	Permutation Importance
(E_{mean})	0.297
(S_{mean})	0.078
(w_{covid})	0.056
<i>others</i>	~ 0

Interpretation

- Elasticity is approximately **4× more influential** than stress.
- Other parameters influence outcomes primarily through their effect on (E) and (S) .

8.8 Phase diagram interpretation

The stress–elasticity plane can be divided into three regions:

1. Household-dominant region

- high elasticity
- low-to-moderate stress

2. Market-dominant region

- low elasticity or high stress

3. Transition band

- intermediate values
- high sensitivity to perturbations

Conceptual interpretation

The system exhibits a **phase-like structure**, where:

- elasticity determines whether alternative routing is possible,
- stress determines whether available capacity is overwhelmed.

8.9 Summary of regime structure

The classifier results establish that:

1. Exit-channel dominance is governed by a **low-dimensional state space**.
2. Elasticity is the **primary control variable**, acting as a necessary condition for household-channel routing.
3. Stress acts as an **overload variable**, forcing market-channel dominance at high levels.
4. The system exhibits a **sloped regime boundary** with a finite-width transition band.

Figure 15. Stress–Elasticity Phase Diagram with Classifier Boundary

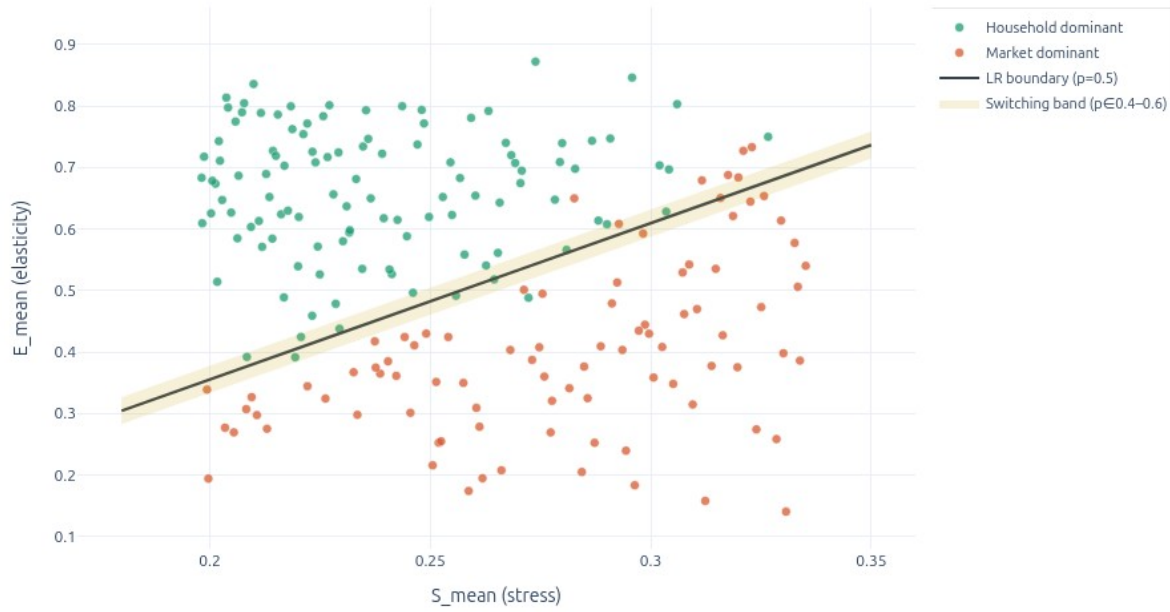


Figure 15: Stress–elasticity phase diagram. Scatter plot of sector-run observations with household vs market dominance, overlaid with classifier boundary.

Figure 16. Logistic Probability Surface — P(Household Dominant) in S–E Space

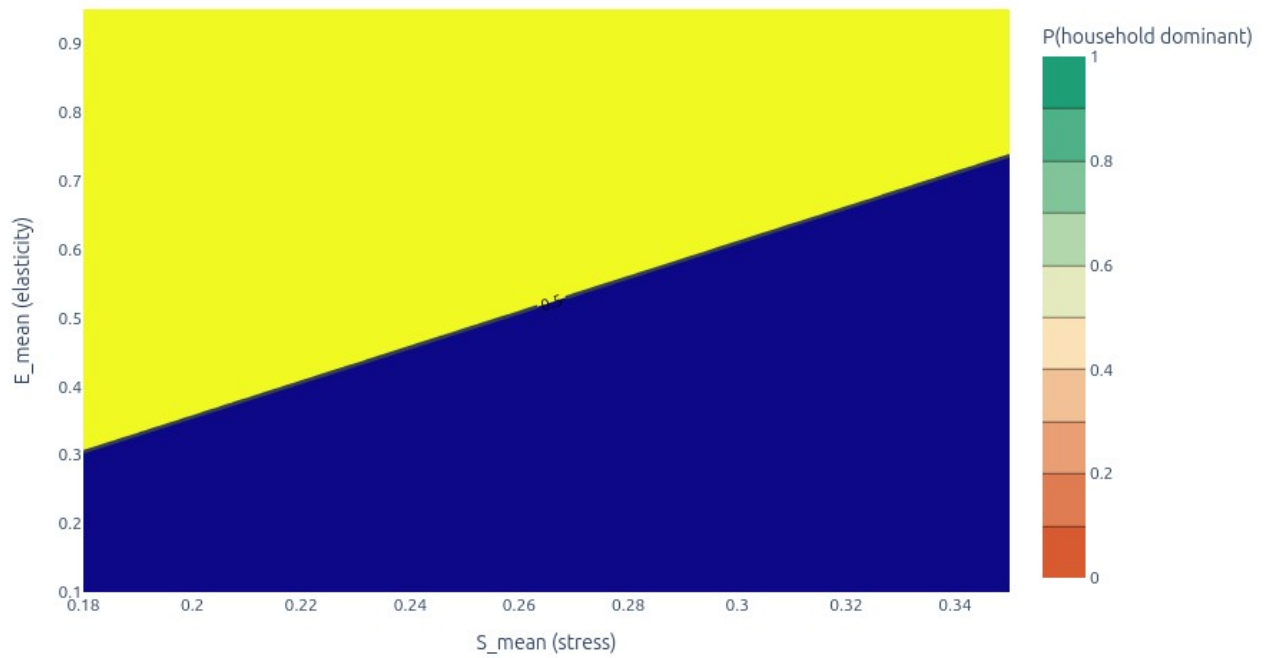


Figure 16: Logistic probability surface. Contour plot showing probability of household dominance across S–E space.

Figure 17. Decision Tree Threshold Overlay on Continuous LR Boundary

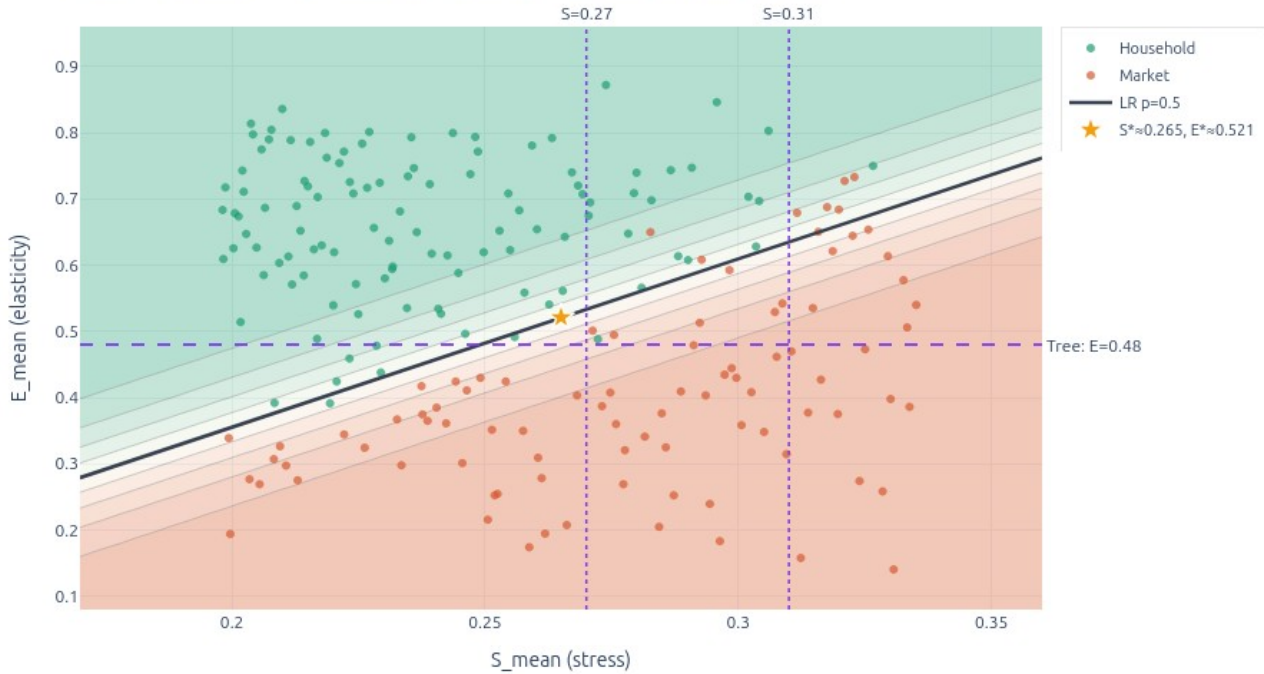


Figure 17: Decision tree boundary overlay. Visualization of discrete thresholds ($E=0.48$, $S=0.27$, $S=0.31$) superimposed on continuous boundary.

9. Discussion and System Interpretation

This section interprets the results of the coupled SER–SIR model, focusing on how epidemic forcing interacts with stress, elasticity, and network structure to produce distinct exit-channel regimes. The emphasis is on mechanism and system behavior rather than additional empirical claims.

9.1 Epidemic shocks as routing forces

A central implication of the coupled model is that epidemic shocks do not merely increase stress levels—they alter how stress is **routed through the system**.

In the SER–SIR framework:

- epidemic prevalence generates a **time-varying pressure field**,
- this pressure accumulates as sectoral stress,
- and the system must determine how to release unresolved burden.

The key result is that this release is not uniform. Instead, stress exits through one of two channels—market or household—depending on system state.

Interpretation

Epidemic shocks should therefore be understood as **routing forces**:

- they push the system toward regions of state space where different exit mechanisms dominate,
- rather than simply increasing the magnitude of observable outcomes.

9.2 Elasticity as the primary control variable

Across all analyses, elasticity emerges as the dominant determinant of exit-channel behavior.

Key role of elasticity

- It acts as a **buffer stock**, absorbing stress while capacity remains.
- It determines whether alternative routing (household channel) is possible.
- It sets a **floor condition**: below a threshold, household-channel leakage disappears.

Implication

The system's ability to avoid visible failure depends primarily on retained elasticity, not on stress levels alone.

Stress may rise significantly without immediate structural failure if elasticity remains sufficient.

9.3 Stress as an overload condition

While elasticity governs the availability of alternative routing, stress determines when available capacity is overwhelmed.

Observed behavior

- At low stress, leakage is minimal.
- At moderate stress, leakage is routed through household channels.
- At high stress, leakage shifts abruptly into market channels.

Interpretation

Stress acts as a **threshold-triggering variable**:

- once it exceeds the capacity supported by elasticity,
- the system is forced into market-visible failure.

9.4 Competing exit channels

The parameter sweep establishes that:

$$[L_{\text{market}} \wedge L_{\text{household}}]$$

are consistently negatively related across the explored domain.

Interpretation

This implies that:

- both channels draw from the same underlying stress budget,
- and the system allocates leakage between them depending on state.

Importantly:

The channels are not sequential (one feeding the other), but **competitive**.

At any point in time, the system preferentially routes stress through one channel or the other.

9.5 The transition regime

The regime boundary analysis reveals a finite-width transition band in stress–elasticity space.

Characteristics of the transition band

- moderate stress levels,
- partially degraded elasticity,
- high sensitivity to perturbations,
- variability in channel dominance.

Interpretation

This region represents a **transition regime** in which:

- the system remains structurally functional,
- but stress is increasingly displaced into household channels,
- and small changes in conditions can trigger a shift to market-dominant leakage.

This regime is critical because it precedes observable structural breakdown.

9.6 Household-channel leakage as a boundary process

The statistical analysis shows that ($L_{household}$):

- is non-Gaussian,
- is zero-inflated,
- activates episodically,
- and exhibits structural breaks.

Interpretation

These properties are consistent with a **boundary exit process**, rather than a continuous internal variable.

This suggests that:

Household-channel leakage represents stress that exits the modeled economic system into human, local, and temporal domains.

It is therefore:

- not directly observable through standard economic metrics,
- and not continuously active.

9.7 Visibility and timing of system degradation

Combining the regime and statistical results leads to an important implication:

- structural failure (market-channel leakage) occurs **after** elasticity has been significantly depleted,
- while household-channel leakage can be active earlier.

Interpretation

This implies that:

Systems may experience substantial internal degradation before this degradation becomes visible through conventional market indicators.

In practical terms:

- stress can be displaced into household domains,
- masking the extent of system strain,
- until the system crosses the regime boundary.

9.8 Relationship to surface observation

The framework also provides a structured interpretation of surface-level observations.

Individuals and households directly experience:

- reduced participation,
- deferral of consumption,
- increased effort to maintain normal activity.

These correspond to the activation of the household-channel leakage.

Interpretation

This suggests that:

Observations made at the household or consumer level may detect system strain earlier than institutional metrics that track market outcomes.

This is not because such observations are inherently more accurate, but because they are **closer to the exit channel through which stress is being routed** during the transition regime.

9.9 The Snapback Illusion

An implication of the two-channel leakage structure is the possibility of what may be termed a **snapback illusion**.

This occurs when observable market indicators suggest stabilization or recovery following a shock, while underlying system stress remains elevated. In the SER–SIR framework, this corresponds to a regime in which:

- market-channel leakage (L_{market}) is low,
- household-channel leakage ($L_{household}$) is active,
- and elasticity (E) remains above the threshold required for household-channel routing.

In this regime, stress is not resolved but **displaced into household and temporal domains**, including deferral, reduced participation, and localized coping mechanisms. Because these processes are not fully captured by standard economic metrics, the system may appear to recover even as internal strain persists.

As elasticity continues to erode, the system may cross the stress–elasticity boundary identified in Section 8. At that point, the capacity to route stress through household channels diminishes, and leakage shifts into market channels. This transition can produce a rapid increase in observable instability, giving the impression of a sudden deterioration following a period of apparent stability.

The snapback illusion therefore reflects a mismatch between:

- **observed system response**, and
- **underlying stress routing**.

It is not a reversal of prior recovery, but the delayed manifestation of stress that was previously displaced outside observable channels.

9.10 System interpretation summary

The coupled SER–SIR system can be interpreted as follows:

1. Epidemic pressure increases sectoral stress.
2. Elasticity buffers stress while capacity remains.

3. As elasticity declines, stress must exit the system.
4. Exit occurs through two competing channels:
 - household (latent),
 - market (observable).
5. The active channel is determined by the joint stress–elasticity state.

Figure 18. Snapback Illustration — Shelter: Apparent Stabilization Followed by Delayed Market-Channel Activation

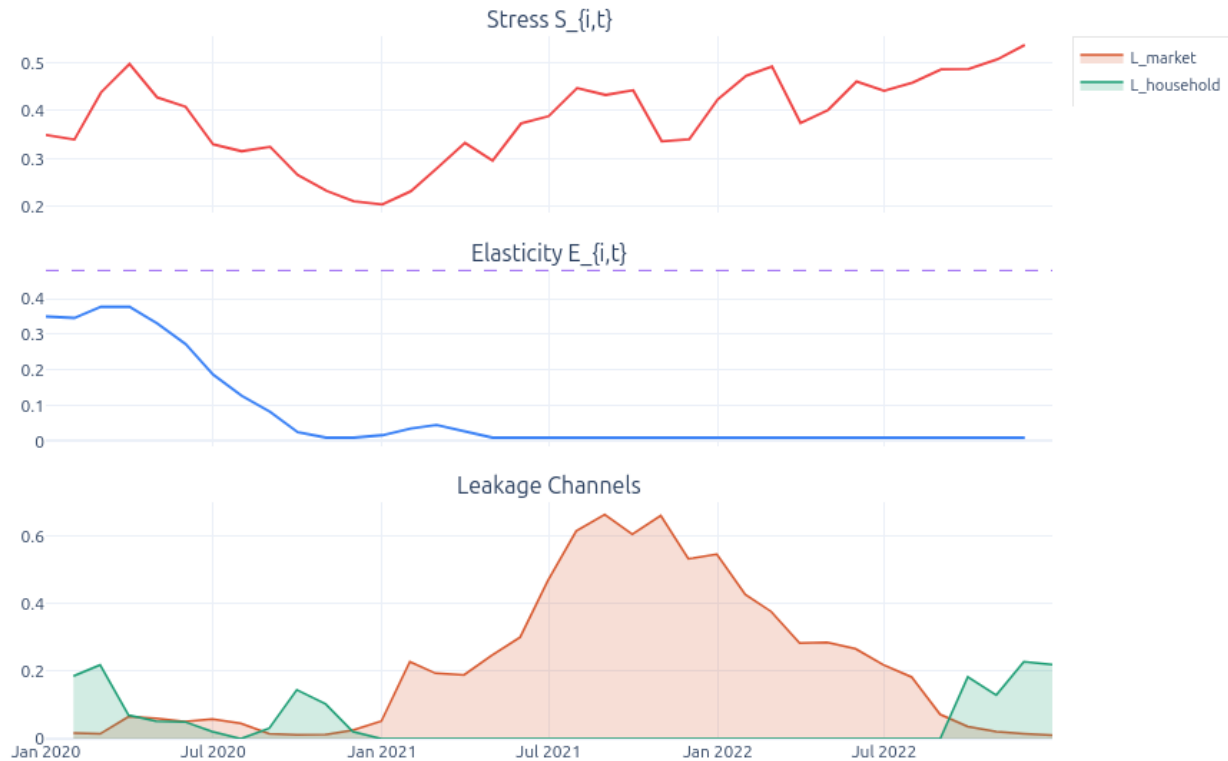


Figure 18: Snapback Illustration

Time series showing stress, elasticity, and leakage channels, illustrating apparent stabilization followed by delayed market-channel activation.

10. Implications and Limitations

This section outlines the implications of the coupled SER–SIR framework for economic analysis and policy, and discusses key limitations of the current approach.

10.1 Implications for economic analysis

10.1.1 Beyond observable market outcomes

A central implication of the model is that observable market outcomes do not fully capture system strain.

Because stress can exit through household channels:

- periods of apparent stability may coexist with substantial underlying burden,
- conventional indicators (prices, output, employment) may lag system conditions,
- and structural degradation may be underestimated prior to regime shifts.

Interpretation

Market-based metrics should be interpreted as **partial observations** of system state, not complete representations.

10.1.2 The role of elasticity

The results indicate that elasticity is the primary determinant of system resilience.

Implications

- Policies or conditions that maintain elasticity can delay or prevent regime transitions.
- Systems with similar stress levels may exhibit very different outcomes depending on elasticity.
- Monitoring elasticity (even indirectly) may provide more information about system stability than stress alone.

10.1.3 Household-channel leakage as an early signal

The activation of household-channel leakage occurs in the transition regime, prior to structural failure.

Implications

- Household-level indicators (e.g., participation, spending patterns, deferral behavior) may provide early signals of system strain.
- These signals are not fully captured in standard economic aggregates.
- Incorporating such indicators may improve detection of emerging instability.

10.1.4 Regime-aware interpretation

The existence of a regime boundary implies that system behavior is state-dependent.

Implications

- Relationships between variables may change across regimes.
- Linear or average-based interpretations may obscure critical transitions.
- Analytical frameworks should account for:
 - threshold behavior,
 - regime switching,

- and nonlinear responses.

10.2 Implications for policy interpretation

10.2.1 Burden displacement vs resolution

Policies that rely on household absorption of stress may not resolve underlying system strain.

Interpretation

- Such policies may **delay visible failure**,
- but do so by shifting burden into less observable domains.

Implication

Distinguishing between **burden resolution** and **burden displacement** is critical.

10.2.2 Timing of interventions

Because regime transitions depend on elasticity:

- early interventions that preserve elasticity may be more effective,
- while late interventions may face systems already in market-dominant regimes.

Implication

Policy effectiveness is **state-dependent**, not only magnitude-dependent.

10.2.3 Monitoring regime position

The stress–elasticity framework suggests that monitoring system position relative to the regime boundary may be useful.

Implication

- Identifying when the system approaches the transition band could inform:
 - targeted support,
 - or preventive measures.

10.3 Limitations

10.3.1 Measurement of household-channel leakage

Household-channel leakage is not directly observed.

- It is inferred as a residual.
- Its interpretation depends on the validity of expected vs observed activity construction.

Implication

Results should be interpreted as **evidence of a consistent residual pattern**, not direct measurement of household behavior.

10.3.2 Data constraints

The model relies on:

- aggregated sector proxies,
- publicly available data,
- and constructed variables.

Implication

- sector definitions are simplified,
- and finer-grained data may refine results.

10.3.3 Model structure

The SER–SIR framework is:

- a reduced-form, state-based model,
- not a full structural general equilibrium system.

Implication

- results demonstrate **mechanism and structure**,
- but do not constitute a complete economic model.

10.3.4 Parameter sweep scope

The parameter sweep explores a broad but finite region of model space.

Implication

- robustness is established within this domain,
- but generalization beyond it requires further testing.

10.3.5 External validity

While motivated by real-world data, the model is not calibrated to match specific empirical outcomes.

Implication

- findings should be viewed as structural insights,
- requiring further validation against detailed empirical datasets.

10.4 Summary

The coupled SER–SIR framework suggests that:

- stress routing is as important as stress magnitude,
- elasticity governs the availability of alternative exit channels,
- and systems may transition between qualitatively different regimes.

At the same time, the model relies on inferred components and simplified structure, requiring careful interpretation and further validation.

11. Conclusion

This paper presents a coupled epidemic–economic framework that models how stress generated by an external shock is absorbed, transmitted, and ultimately released within a sectoral system. By integrating SIR epidemic dynamics with a Stress–Elasticity–Reaction (SER) economic model, the analysis moves beyond treating shocks as one-time disturbances and instead represents them as time-varying pressure fields interacting with system capacity.

A central contribution of the framework is the explicit decomposition of leaked stress into two exit channels: a **market channel**, through which stress becomes visible in observable economic outcomes, and a **household channel**, through which stress is propagated into human, local, and temporal domains outside standard measurement. The statistical behavior of household-channel leakage—non-Gaussian, zero-inflated, and episodic—indicates that it functions as an intermittently activated boundary process rather than a continuous internal variable.

A second contribution is the demonstration that these two channels operate as **competing exit routes**. Across a broad parameter sweep, market-channel and household-channel leakage are consistently negatively related, with no observed sign reversals. This establishes that the system allocates unresolved stress between alternative pathways rather than expressing it uniformly.

The third and most important result is the identification of a **regime boundary** in stress–elasticity space governing exit-channel dominance. Classification results show that this boundary is low-dimensional and primarily determined by mean stress and mean elasticity. Elasticity acts as a necessary condition for household-channel routing, while stress acts as an overload condition that forces market-channel dominance. The resulting boundary is sloped and exhibits a finite-width transition band, within which small changes in system state can shift the dominant exit channel.

Taken together, these results imply that economic systems may undergo substantial internal degradation before this degradation becomes visible through conventional market indicators. In particular, stress can be displaced into household channels during a transition regime, delaying observable structural failure. This suggests that analyses relying solely on market-based signals may underestimate the extent and timing of system strain.

The framework also highlights the importance of elasticity as a control variable. Maintaining absorptive capacity appears central to preventing regime transitions, while depletion of elasticity

constrains the system's ability to route stress through non-market channels. This provides a mechanism linking external shocks, internal capacity, and observable outcomes.

Several limitations remain. Household-channel leakage is inferred rather than directly measured, and the model relies on aggregated sector proxies and a reduced-form structure. The robustness of the results is established within the explored parameter space but requires further validation against detailed empirical datasets. Future work may extend the framework by incorporating higher-resolution data, refining activity measurement, and testing the regime boundary against observed economic behavior.

Despite these limitations, the results suggest that modeling stress routing and exit-channel competition provides a useful perspective for understanding how systems respond to large-scale shocks. By focusing on the interaction between stress and capacity, rather than on outcomes alone, the coupled SER–SIR framework offers a structured approach to analyzing both the timing and visibility of system degradation.

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